

Ubaid Azam

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RESEARCH PROFILE

Researcher in **trustworthy and responsible AI**: knowing when a model's output can be relied on, and showing why. I develop Bayesian methods that attach calibrated uncertainty to predictions in safety-critical and low-resource domains, including clinical and legal text and under-represented languages; I evaluate explanations for faithfulness rather than assuming them; and I adapt large language models to specialist domains through prompt tuning, including federated settings where data cannot be pooled. Ten peer-reviewed papers, seven as first author, at venues including WSDM, CIKM, COLING and LREC. Primary researcher on PICASO, a competitively awarded Alan Turing Institute grant, and led a team in the United Nations Development Programme's AI Trust and Safety Re-Imagination Programme. Winner of University of Southampton Director's Awards for Public Engagement and Knowledge Exchange.

EDUCATION

University of Southampton, United Kingdom PhD, Computer Science. Thesis: Trustworthy NLP for Low-Resource Domains.	Sep 2023 – Present
Lahore University of Management Sciences, Pakistan MS, Computer Science; CGPA 3.59 / 4.0	Jul 2019 – Sep 2021
University of Gujrat, Pakistan BS (Hons.) Software Engineering; CGPA 3.90 / 4.0, Gold Medallist	Sep 2014 – Sep 2018

PUBLICATIONS

- PROCS: Uncertainty-Aware Knowledge Graph Embeddings via Conceptual Spaces.
U. Azam, I. Razzak, S. Schockaert, S. Jameel. *CIKM 2026* (to appear).
- Towards Explainable Hate Speech Detection in Roman Urdu. ↗
U. Azam, H. Rizwan, F. Ullah, A. Faheem, F. Kamiran, A. Karim. *IJCNN 2026*.
- Lightweight Cross-Lingual Federated Prompt Tuning for Low-Resource Languages. ↗
U. Azam, I. Razzak, S. Jameel. *LREC 2026*.
- BAKER: Bayesian Kernel Uncertainty in Domain-Specific Document Modelling. ↗
U. Azam, I. Razzak, S. Vishwakarma, H. Hacid, D. Zhang, S. Jameel. *WSDM 2025*.
- Uncertainty Modelling in Under-Represented Languages with Bayesian Deep Gaussian Processes. ↗
U. Azam, I. Razzak, S. Vishwakarma, S. Jameel. *COLING 2025*.
- Part of Speech (POS) Tagging in Roman Urdu: Datasets and Models. ↗
A. Faheem, F. Ullah, U. Azam, M.S. Ayub, A. Karim. *Language Resources and Evaluation*, 59(4), 2025.
- Would You Trust an AI Doctor? Building Reliable Medical Predictions. ↗
U. Azam, I. Razzak, S. Vishwakarma, H. Hacid, D. Zhang, S. Jameel. *WISE 2024*.
- Detecting Cybercrimes in Accordance with Pakistani Law: Dataset and Evaluation using PLMs. ↗
F. Ullah, A. Faheem, U. Azam, M.S. Ayub, F. Kamiran, A. Karim. *LREC-COLING 2024*.
- Comparing Prompt-Based and Standard Fine-Tuning for Urdu Text Classification. ↗
F. Ullah, U. Azam, A. Faheem, F. Kamiran, A. Karim. *Findings of EMNLP 2023*.
- Exploring Data Augmentation Strategies for Hate Speech Detection in Roman Urdu. ↗
U. Azam, H. Rizwan, A. Karim. *LREC 2022*.

Manuscripts Under Review

- Adaptive and Resource-Aware Federated Prompt Learning for Heterogeneous Healthcare.
U. Azam, I. Razzak, S. Jameel. *Scientific Reports (Nature)*, under review.

RESEARCH PROGRAMMES & GRANTS

Alan Turing Institute Grant: PICASO ↗

Dec 2025 – Present

Knowledge Graphs and Ontologies

- Primary researcher on this competitively awarded grant, *PICASO: Probabilistic Conceptual Spaces Sensemaking under Uncertainty* (PI: Dr Shoaib Jameel); contributed to writing the proposal.
- Currently extending the work to fuse text-derived and graph-derived evidence so that disagreement between sources is kept as measurable uncertainty rather than averaged away.
- Built the **Defence-Wikidata** dataset by writing a multi-stage extraction pipeline over a full Wikidata dump: streaming the full compressed dump, selecting entities by class hierarchy, following two-hop relations, and validating and normalising the output into 68,826 entities and 143,377 triples across 355 relation types.
- Developed **PROCS** ↗, which attaches calibrated uncertainty to knowledge graph predictions so that a generated fact carries a signal of how far it can be relied on (CIKM 2026).
- Benchmarked against FB15k-237 and WN18RR, and **released the code, the dataset and the canonical split under an open licence** so the results can be reproduced independently.
- Report progress to an international project team through regular technical summaries and meetings.

United Nations AI Trust and Safety Re-Imagination Programme ↗

Sep 2025 – Jan 2026

- Led research team **TrustWeave** developing trustworthy AI solutions for under-represented languages.
- Collaborated with global experts across technical, legal, and policy domains.
- Presented work to UN mentors and partner organisations (**featured by UNDP Digital** ↗).

PROFESSIONAL EXPERIENCE

Research Assistant, *University of Southampton*

Dec 2025 – Present

- Employed on PICASO, an Alan Turing Institute grant (PI: Dr Shoaib Jameel), as the project's primary researcher; see Research Programmes & Grants above.

Master's Project Mentor, *University of Southampton, ECS*

2024, 2025, 2026

- Mentored **around 30 Master's students** across three consecutive years, from project scoping and choice of methods through to evaluation and writing up.
- Advised on experimental design and on reporting results honestly, including what to do when a result does not hold.

Laboratory Demonstrator, *University of Southampton, ECS*

2024 – Present

- Teach the practical component of the Machine Learning and Deep Learning modules to cohorts of around 200.

Lecturer, *The University of Lahore*

Feb 2022 – Jul 2023

- Designed and delivered undergraduate modules in Introduction to Machine Learning, Software Engineering and Software Project Management, to cohorts of 50 to 200 students.
- Supervised final-year student projects.

Research Assistant, *Lahore University of Management Sciences*

Jul 2021 – Jul 2023

- Applied large language models and prompt-based methods to text classification, comparing prompting against standard fine-tuning.
- Used generative models including GPT-2 to synthesise training data for low-resource languages.

- Built and annotated corpora for specialist domains, including a cybercrime dataset labelled against definitions drawn from Pakistani law.

Graduate Teaching Assistant, *LUMS, Topics in Data Mining*

Sep 2021 – Dec 2021

- Assisted in preparing lesson plans for graduate and postgraduate students.
- Presented research papers to the class.

College Teaching Intern, *Govt. Graduate College of Science*

Nov 2018 – Aug 2019

- Taught students at BS level; maintained and invigilated classes.

HONOURS & AWARDS

- **Winner**, Doctoral College Director's Award for Public Engagement & Outreach and Knowledge Exchange & Enterprise, University of Southampton (2026).
- **Winner**, research publication recognition, Doctoral College Director's Awards, University of Southampton (2026).
- Nominated, Doctoral College Director's Award for Citizenship and Community, University of Southampton (2025 & 2026, two consecutive years).
- Gold Medallist, BS (Hons.) Software Engineering.

TECHNICAL SKILLS

Uncertainty	Bayesian & probabilistic deep learning, deep Gaussian processes, calibration, selective prediction, flagging low-confidence predictions for human review
Responsible AI	Explanation methods and their <i>evaluation</i> for plausibility and faithfulness; robustness under adversarial variation and temporal distribution shift; privacy-preserving federated learning
LLMs	Prompt engineering and prompt tuning for domain-specific applications, parameter-efficient and federated fine-tuning, generative data synthesis, controlled comparison of prompting against fine-tuning
Knowledge graphs	Uncertainty-aware knowledge graph embeddings, conceptual spaces, large-scale extraction pipelines over Wikidata
Domains	Clinical text (BioBERT, BlueBERT, ClinicalBERT), legal text, online harms, under-represented languages
Data	Dataset construction and annotation to externally defined guidelines (hate speech, cyber-crime law)
ML / DL	PyTorch, TensorFlow, Hugging Face Transformers, GPyTorch (probabilistic ML)
Languages	Python, C++, Java, SQL; NumPy, pandas, SciPy, scikit-learn
Evaluation	Benchmark design, fixed reproducible splits, repeated-run and variance reporting, ablation studies
Tools	LaTeX, Git, Linux; GPU training (NVIDIA RTX 6000 Ada)

SERVICE & OUTREACH

- Invited speaker, Learning & Reasoning Seminar, Cardiff University (Computer Science), Apr 2026.
- Reviewer, CIKM and IEEE IJCNN.
- Release code and datasets with my publications (github.com/ubaidazam ↗) so results can be reproduced and reused.
- Designed and delivered sessions at the AI and Society International Summer School, University of Southampton (2025), for students from computer science, law, policy and the social sciences.
- Helped organise the ECS PhD recruitment event, University of Southampton (2025); managed logistics and applicant queries.